Reg. No.														

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, NOVEMBER 2025

OPEN BOOK EXAMINATION

Fourth Semester

21ECC204T - SIGNAL PROCESSING

(For the candidates admitted during the academic year 2021 - 2022 to 2024 - 2025)

Note:

i. Specific approved THREE text books (Printed or photocopy) recommended for the course.
ii. Handwritten class notes (certified by the faculty handling the course / Head of the Department).

Time: Three Hours **Max. Marks: 75**

INSTRUCTIONS

Answer **ANY FIVE** Questions from **Q.No: 1 to 7**
(Q.No 8 is compulsory)

Marks BL CO

PART - A (5 × 10 = 50 Marks)

Answer **any 5** Questions

		Marks	CO	BL
1.	Find out whether the following signals are energy or power signal or neither power nor energy as the case may be for the signal. (i) $X(t) = u(t) + 5u(t-1) - 2u(t-2)$ (ii) $X(t) = \cos(2t) + \sin(t/5)$	10	1	3
2.	Using the suitable property determine the Fourier Transform of the signal $x(t) = sin\Omega_0(t)cos\Omega_0(t)$	10	2	3
3.	An LTI system is described by the following differential equation: $\frac{dy(t)}{dt} + 5y(t) = x(t)$. Determine the output y(t) using Fourier Transform for the following input signals: (i) $e^{-2t}u(t)$ (ii) $x(t) = \delta(t)$	10	2	3
4.	Find 8-point DFT of the sequence (using DFT formula) $x(n) = \{1,1,1,1\}$	10	3	3
5.	Perform Linear Convolution of the given sequence $x(n)=\{1, 2, 0.5, 1\}$; $h(n)=\{1, 2, 1, -1\}$	10	3	3
6.	Design an FIR filter for the ideal frequency response using Hamming window with N=7 $H_d(e^{j\omega}) = e^{-j3\omega}$; $-\pi/8 < w < \pi/8$ 0 ; $\pi/8 < w < \pi$	10	4	3
7.	The analog transfer function $H(s) = \dfrac{2}{(s+1)(s+2)}$. Determine H(z) using impulse invariance method.	10	5	3

PART - B (1 × 25 = 25 Marks)

Answer **all** Question

Marks CO BL

8. a) (i) Design a digital Butterworth filter satisfying the constraints using bilinear transformations. 25 3 3

$0.707 \leq |H(\omega)| \leq 1.0 \; ; \; 0 \leq \omega \leq \pi/2$

$|H(\omega)| \leq 0.2 \; ; \; 3\pi/4 \leq \omega \leq \pi.$

(ii) Design a digital Butterworth filter satisfying the constraints

$0.8 \leq |H(\omega)| \leq 1.0 \; ; \; 0 \leq \omega \leq \pi/4$

$|H(\omega)| \leq 0.2 \; ; \; \pi/2 \leq \omega \leq \pi.$

Apply Bilinear transformation method.

(OR)

b) (i) Compute the DFT of the sequence $x(n)=cos\left(\frac{n\pi}{2}\right)$ where N=4 using DIF FFT 25 3 3

(ii) Find IDFT of the sequence using DIT-FFT

X(k)= {6, -2+j2, -2, -2-j2}

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